CAS SciFinder

Task History

Initiating Search

November 10, 2025, 2:07 PM

Search:

CAS Lexicon Search:

Barton deoxygenation - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (191)	**References**	View Results
Exported: Retrieved Related Reaction Results + Filters (70) Filtered By: Yield: **90-100%, 80-89%, 70-79%** Document Type: **Journal**	**Reactions**	View Results

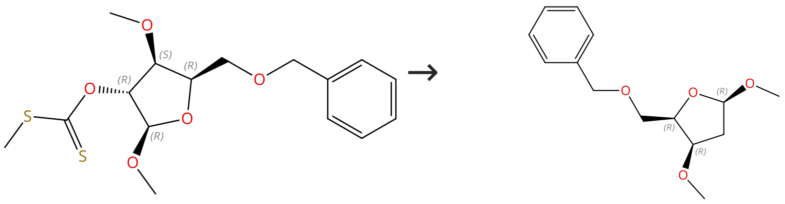
CAS SciFinder

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction)

Steps: **1** Yield: **100%**



Absolute stereochemistry shown, Rotation (-)

Absolute stereochemistry shown, Rotation (-)

Supplier (1)

31-535-CAS-5309335

Steps: **1** Yield: **100%**

1.1 **Reagents:** Tributylstannane
Catalysts: 1,1'-Azobis(cyclohexane-1-carbonitrile)
Solvents: Benzene; 80 °C; 2 h, 80 °C

Experimental Protocols

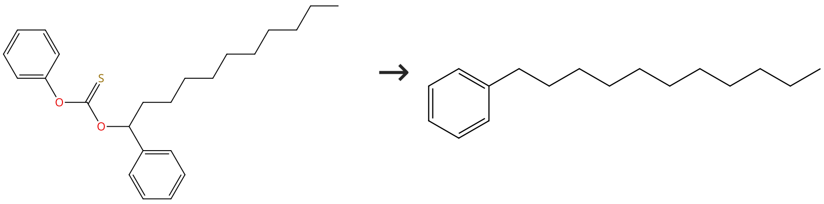
β-Oxygen Effect in the Barton-McCombie Deoxygenation Reaction: Further Experimental and Theoretical Findings

By: Sanchez-Eleuterio, Alma; et al

Journal of Organic Chemistry (2013), 78(18), 9127-9136.

Scheme 2 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (47)

31-535-CAS-6527993

Steps: **1** Yield: **100%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Cyclopentyl methyl ether; 30 min, 90 °C; 1 h, 90 °C

Experimental Protocols

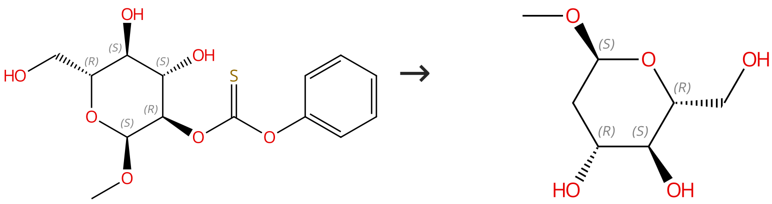
Evaluation of cyclopentyl methyl ether (CPME) as a solvent for radical reactions

By: Kobayashi, Shoji; et al

Tetrahedron (2013), 69(10), 2251-2259.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **99%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown, Rotation (+)

Supplier (1)

Supplier (1)

31-535-CAS-14301282 Steps: **1** Yield: **99%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 4 h, reflux

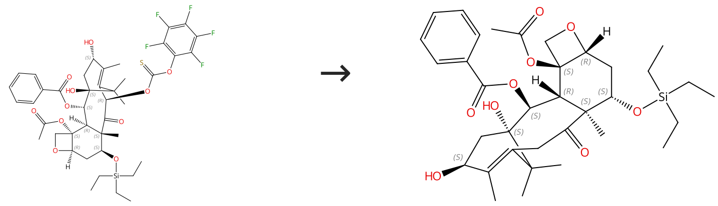
Experimental Protocols

Organotin-Catalyzed Highly Regioselective Thiocarbonylation of Non-Protected Carbohydrates and Synthesis of Deoxy Carbohydrates in a Minimum Number of Steps

By: Muramatsu, Wataru; et al

Chemistry - A European Journal (2012), 18(16), 4850-4853, S4850/1-S4850/63.

Scheme 4 (1 Reaction) Steps: **1** Yield: **99%**



Absolute stereochemistry shown

Absolute stereochemistry shown

31-535-CAS-736334 Steps: **1** Yield: **99%**

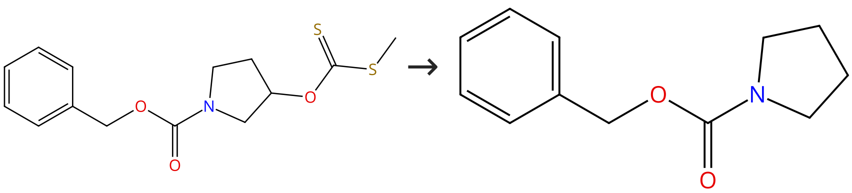
1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene

Synthesis of 7-deoxy- and 7,10-dideoxytaxol via radical intermediates

By: Chen, Shu Hui; et al

Journal of Organic Chemistry (1993), 58(19), 5028-9.

Scheme 5 (1 Reaction) Steps: **1** Yield: **98%**



Suppliers (55)

31-535-CAS-22855827 Steps: **1** Yield: **98%**

1.1 **Reagents:** Tris(trimethylsilyl)silane
Catalysts: Chlorophyll a
Solvents: Dimethyl sulfoxide; 5 h, 50 - 55 °C

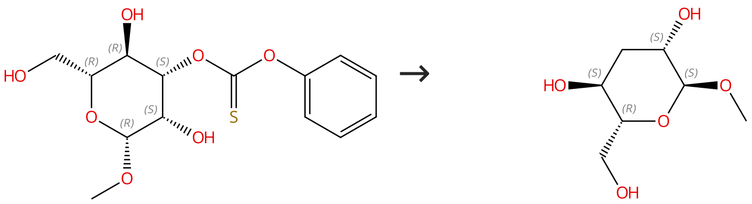
Experimental Protocols

Red-Light-Mediated Barton-McCombie Reaction

By: Ogura, Akihiro; et al

Bulletin of the Chemical Society of Japan (2020), 93(7), 936-941.

Scheme 6 (1 Reaction) Steps: **1** Yield: **98%**



Absolute stereochemistry shown, Rotation (-)

Absolute stereochemistry shown, Rotation (+)

Supplier (1)

Suppliers (4)

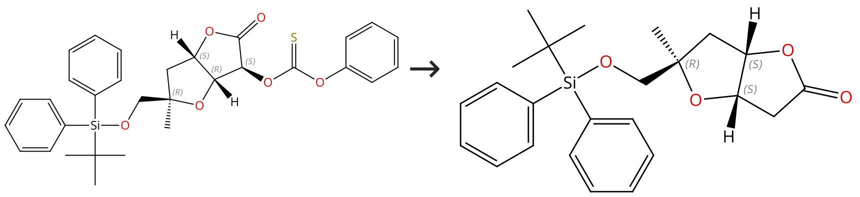
31-535-CAS-10317632	Steps: **1** Yield: **98%**
1.1	**Reagents:** Tributylstannane **Catalysts:** Azobisisobutyronitrile **Solvents:** Toluene; 4 h, reflux
Experimental Protocols	

Organotin-Catalyzed Highly Regioselective Thiocarbonylation of Non-Protected Carbohydrates and Synthesis of Deoxy Carbohydrates in a Minimum Number of Steps

By: Muramatsu, Wataru; et al

Chemistry - A European Journal (2012), 18(16), 4850-4853, S4850/1-S4850/63.

Scheme 7 (1 Reaction) Steps: **1** Yield: **97%**



Absolute stereochemistry shown

Absolute stereochemistry shown, Rotation (-)

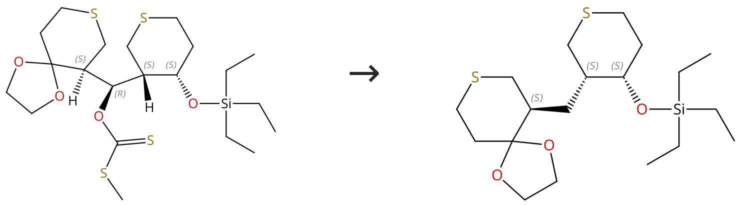
31-535-CAS-13635435	Steps: **1** Yield: **97%**
1.1	**Reagents:** Azobisisobutyronitrile, Tributylstannane **Solvents:** Toluene; 30 min, reflux
Experimental Protocols	

Stereoselective Formation of Tetrahydrofuran Rings via [3 + 2] Annulation: Total Synthesis of Plakortone L

By: Sugimura, Hideyuki; et al

Organic Letters (2014), 16(12), 3384-3387.

Scheme 8 (1 Reaction) Steps: **1** Yield: **97%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown, Rotation (-)

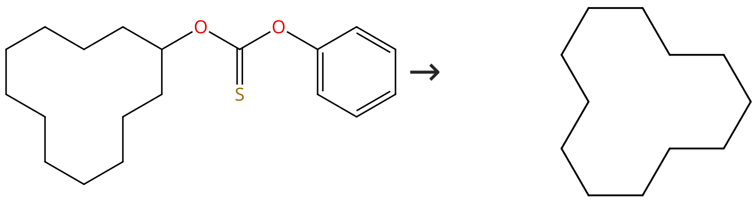
31-535-CAS-11698750	Steps: **1** Yield: **97%**
1.1	**Reagents:** Tributylstannane **Solvents:** Toluene; rt → reflux
1.2	**Catalysts:** Azobisisobutyronitrile; 30 min, reflux; reflux → rt
Experimental Protocols	

Thiopyran Route to Polypropionates: An Efficient Synthesis of Serricornin

By: Ward, Dale. E.; et al

Journal of Organic Chemistry (2006), 71(23), 8989-8992.

Scheme 9 (2 Reactions) Steps: **1** Yield: **92-96%**



Suppliers (2)

Suppliers (56)

31-535-CAS-2263983	Steps: **1** Yield: **96%**
1.1	**Reagents:** Tributylstannane **Catalysts:** 2,2'-Azobis[2,4-dimethylvaleronitrile] **Solvents:** Tetrahydrofuran; 30 min, 70 °C; 3 h, 70 °C
Experimental Protocols	

Evaluation of cyclopentyl methyl ether (CPME) as a solvent for radical reactions

By: Kobayashi, Shoji; et al

Tetrahedron (2013), 69(10), 2251-2259.

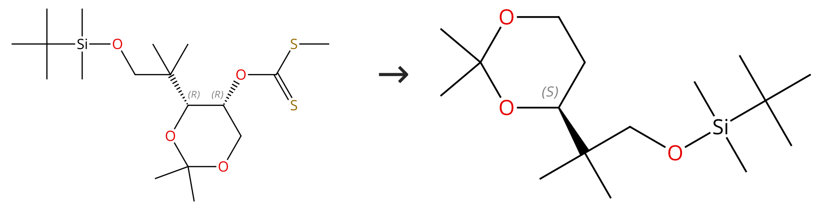
31-535-CAS-3481174	Steps: **1** Yield: **92%**
1.1	**Reagents:** 1,5-Dimethoxy-6-methyl-6-[tris(1-methylethyl)silyl]-1,4-cyclohexadiene **Catalysts:** Azobisisobutyronitrile **Solvents:** Hexane

Silylated cyclohexadienes: new alternatives to tributyltin hydride in free radical chemistry

By: Studer, Armido; et al

Angewandte Chemie, International Edition (2000), 39(17), 3080-3082.

Scheme 10 (1 Reaction) Steps: **1** Yield: **94%**



Absolute stereochemistry shown, Rotation (-) → Absolute stereochemistry shown, Rotation (+)

31-535-CAS-16461097	Steps: **1** Yield: **94%**
1.1	**Reagents:** Tributylstannane **Catalysts:** Azobisisobutyronitrile **Solvents:** Toluene; rt → 80 °C; 3 h, 80 °C

Experimental Protocols

Transition-metal-free stereoselective synthesis of C(1)-C(6) fragment of epothilones and their structural analogues

By: Shklyaruck, Denis G.; et al

Tetrahedron (2016), 72(49), 8015-8021.

Scheme 11 (1 Reaction) Steps: **1** Yield: **94%**

Absolute stereochemistry shown, Rotation (+) → Absolute stereochemistry shown, Rotation (+)

31-535-CAS-1255587	Steps: **1** Yield: **94%**
1.1	**Reagents:** Azobisisobutyronitrile, Tributylstannane **Solvents:** Toluene; 1 h, reflux; 1 h, reflux; reflux → rt

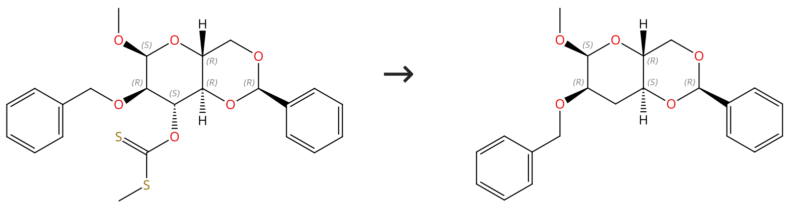
Experimental Protocols

A direct organocatalytic entry to selectively protected aldopentoses and derivatives

By: Grondal, Christoph; et al

Advanced Synthesis & Catalysis (2007), 349(4+5), 694-702.

Scheme 12 (1 Reaction) Steps: **1** Yield: **93%**



Absolute stereochemistry shown → Absolute stereochemistry shown

31-535-CAS-22855507	Steps: **1** Yield: **93%**
1.1	**Reagents:** Tris(trimethylsilyl)silane **Catalysts:** Chlorophyll a **Solvents:** Dimethyl sulfoxide; 5 h, 50 - 55 °C

Experimental Protocols

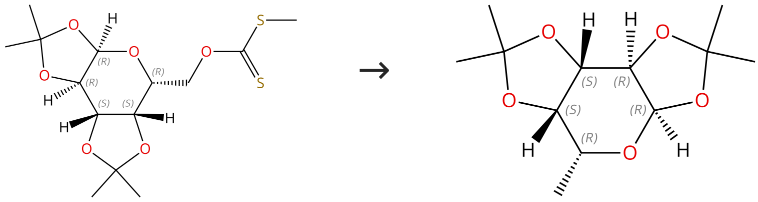
Red-Light-Mediated Barton-McCombie Reaction

By: Ogura, Akihiro; et al

Bulletin of the Chemical Society of Japan (2020), 93(7), 936-941.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **93%**



Absolute stereochemistry shown

Absolute stereochemistry shown, Rotation (-)

Suppliers (42)

31-533-CAS-8706659

Steps: **1** Yield: **93%**

1.1 **Reagents:** Azobisisobutyronitrile, Tris(trimethylsilyl)silane
Solvents: Toluene; 5 min, 130 °C

Experimental Protocols

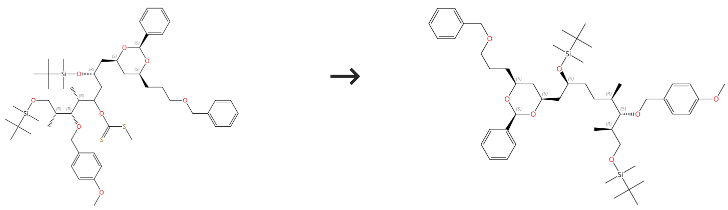
Safe, facile radical-based reduction and hydrosilylation reactions in a microreactor using tris(trimethylsilyl)silane

By: Odedra, Arjan; et al

Chemical Communications (Cambridge, United Kingdom) (2008), (26), 3025-3027.

Scheme 14 (1 Reaction)

Steps: **1** Yield: **92%**



Absolute stereochemistry shown

Absolute stereochemistry shown

31-535-CAS-12997865

Steps: **1** Yield: **92%**

1.1 **Reagents:** Azobisisobutyronitrile, Tributylstannane
Solvents: Toluene; reflux

Experimental Protocols

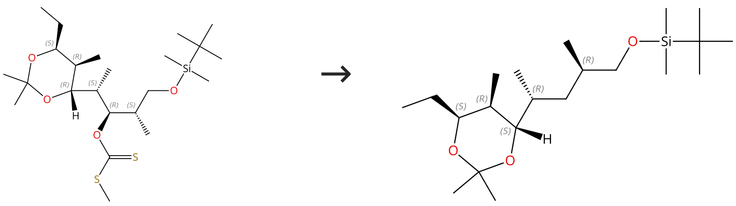
Enantioselective synthesis of the C1-C15 fragment of dolabelide C

By: Vincent, Aurelie; et al

Synlett (2006), (14), 2269-2271.

Scheme 15 (1 Reaction)

Steps: **1** Yield: **92%**



Absolute stereochemistry shown

Absolute stereochemistry shown

31-535-CAS-4009024

Steps: **1** Yield: **92%**

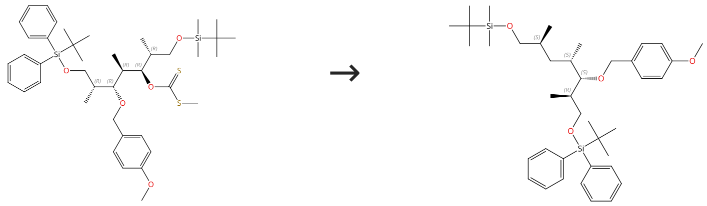
1.1 **Reagents:** Azobisisobutyronitrile, Tributylstannane
Solvents: Benzene; 3 h, 80 °C

Synthesis of C15-C27 segment of venturicidine X by utilizing desymmetrization protocol

By: Yadav, J. S.; et al

Tetrahedron Letters (2010), 51(32), 4179-4181.

Scheme 16 (1 Reaction) — Steps: **1** Yield: **92%**



Absolute stereochemistry shown, Rotation (+) — Absolute stereochemistry shown, Rotation (+)

31-535-CAS-20460140 — Steps: **1** Yield: **92%**

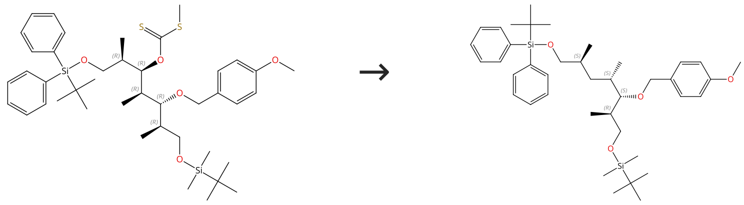
1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 6 h, 80 °C

Studies towards the Synthesis of Thermolide-6'

By: Satyanarayana, Vavilapalli; et al

ChemistrySelect (2018), 3(4), 1000-1003.

Scheme 17 (1 Reaction) — Steps: **1** Yield: **92%**



Absolute stereochemistry shown — Absolute stereochemistry shown

31-535-CAS-19449236 — Steps: **1** Yield: **92%**

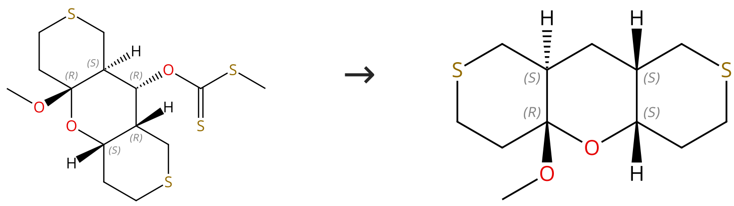
1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; reflux

Stereoselective synthesis of C12-C21 common fragment of thermolides 1-5

By: Satyanarayana, Vavilapalli; et al

Tetrahedron Letters (2018), 59(29), 2828-2830.

Scheme 18 (1 Reaction) — Steps: **1** Yield: **91%**



Absolute stereochemistry shown, Rotation (-) — Absolute stereochemistry shown, Rotation (-)

31-535-CAS-701289 — Steps: **1** Yield: **91%**

1.1 **Reagents:** Tributylstannane
Solvents: Toluene; rt → reflux

1.2 **Catalysts:** Azobisisobutyronitrile; 30 min, reflux; reflux → rt

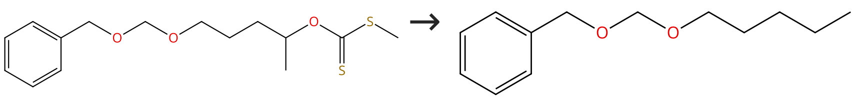
Experimental Protocols

Thiopyran Route to Polypropionates: An Efficient Synthesis of Serricornin

By: Ward, Dale. E.; et al

Journal of Organic Chemistry (2006), 71(23), 8989-8992.

Scheme 19 (1 Reaction) — Steps: **1** Yield: **91%**



Supplier (1)

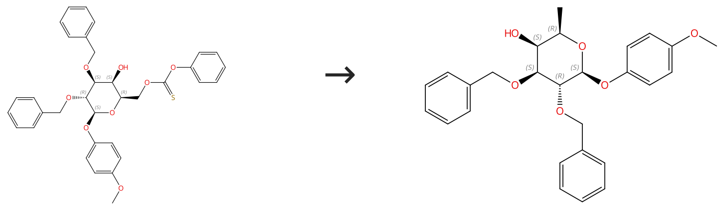
31-535-CAS-22858326	Steps: **1** Yield: **91%**
1.1	**Reagents:** Tris(trimethylsilyl)silane **Catalysts:** Chlorophyll a **Solvents:** Dimethyl sulfoxide; 5 h, 50 - 55 °C
Experimental Protocols	

Red-Light-Mediated Barton-McCombie Reaction

By: Ogura, Akihiro; et al

Bulletin of the Chemical Society of Japan (2020), 93(7), 936-941.

Scheme 20 (1 Reaction) Steps: **1** Yield: **91%**



Absolute stereochemistry shown, Rotation (-) → Absolute stereochemistry shown, Rotation (+)

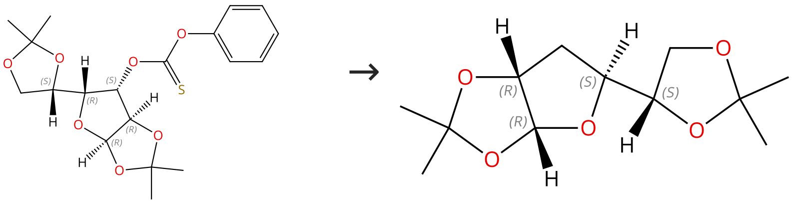
31-533-CAS-22488198	Steps: **1** Yield: **91%**
1.1	**Reagents:** Tris(trimethylsilyl)silane **Catalysts:** Azobisisobutyronitrile **Solvents:** Toluene; 0 °C; 1 h, 80 °C
Experimental Protocols	

Indirect synthetic route to α-L-fucosides via highly stereoselective construction of α-L-galactosides followed by C6-deoxygenation

By: Tomida, Hirotaka; et al

Organic & Biomolecular Chemistry (2020), 18(26), 5017-5033.

Scheme 21 (1 Reaction) Steps: **1** Yield: **91%**



Absolute stereochemistry shown → Absolute stereochemistry shown, Rotation (+)

Suppliers (9)

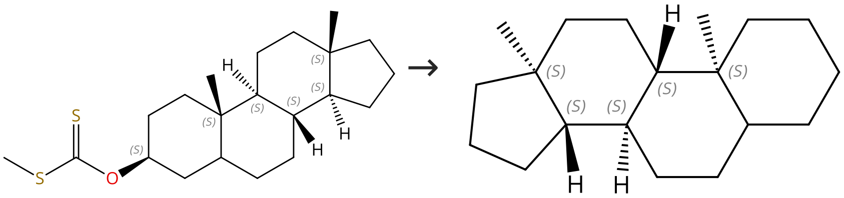
31-535-CAS-5609015	Steps: **1** Yield: **91%**
1.1	**Reagents:** 6-[(1,1-Dimethylethyl)dimethylsilyl]-1,5-dimethoxy-6-methyl-1,4-cyclohexadiene **Catalysts:** Azobisisobutyronitrile **Solvents:** Hexane

Silylated cyclohexadienes: new alternatives to tributyltin hydride in free radical chemistry

By: Studer, Armido; et al

Angewandte Chemie, International Edition (2000), 39(17), 3080-3082.

Scheme 22 (1 Reaction) Steps: **1** Yield: **90%**



Absolute stereochemistry shown → Absolute stereochemistry shown

Suppliers (6)

31-535-CAS-14810658 Steps: **1** Yield: **90%**

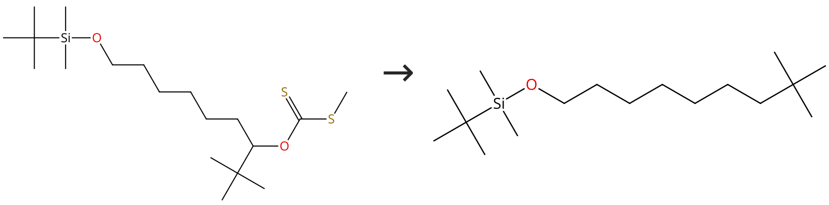
1.1 **Reagents:** Triphenylsilane
Catalysts: 1,1,1-Triphenylsilanethiol, Di-*tert*-butyl hyponitrite
Solvents: 1,4-Dioxane

Formation of silanethiols by reaction of silanes with carbonyl sulfide: implications for radical-chain reduction of thiocarbonyl compounds by silanes

By: Cai, Y.; et al

Tetrahedron Letters (2001), 42(4), 763-766.

Scheme 23 (1 Reaction) Steps: **1** Yield: **90%**



31-535-CAS-22857860 Steps: **1** Yield: **90%**

1.1 **Reagents:** Tris(trimethylsilyl)silane
Catalysts: Chlorophyll a
Solvents: Dimethyl sulfoxide; 5 h, 50 - 55 °C

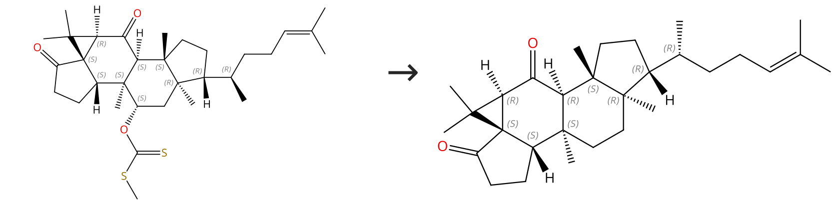
Experimental Protocols

Red-Light-Mediated Barton-McCombie Reaction

By: Ogura, Akihiro; et al

Bulletin of the Chemical Society of Japan (2020), 93(7), 936-941.

Scheme 24 (1 Reaction) Steps: **1** Yield: **90%**



Absolute stereochemistry shown, Rotation (-)

Absolute stereochemistry shown, Rotation (-)

31-614-CAS-44831076 Steps: **1** Yield: **90%**

1.1 **Reagents:** Tris(trimethylsilyl)silane, Red 21
Solvents: Acetonitrile; 12 h, rt

1.2 **Reagents:** Water; rt

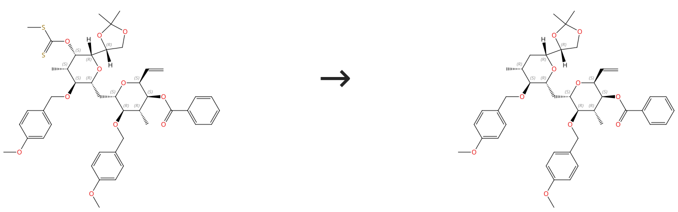
Experimental Protocols

Bioinspired Synthesis of Cucurbalsaminones B and C

By: Zhang, Mengqing; et al

Angewandte Chemie, International Edition (2025), 64(4), e202417318.

Scheme 25 (1 Reaction) Steps: **1** Yield: **90%**



Absolute stereochemistry shown

Absolute stereochemistry shown, Rotation (-)

31-535-CAS-12225669 Steps: **1** Yield: **90%**

1.1 **Reagents:** Triphenylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Benzene; 6 h, reflux

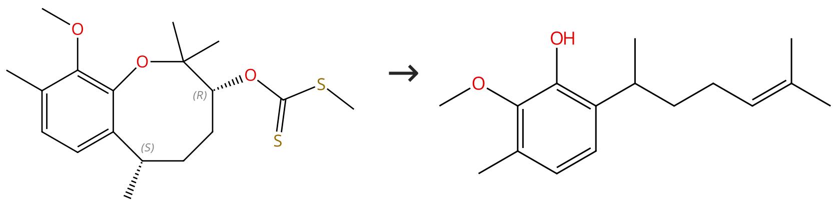
Experimental Protocols

Synthetic Efforts toward the C22-C36 Subunit of Halichondrin B Utilizing Local and Imposed Symmetry

By: Keller, Valerie A.; et al

Organic Letters (2005), 7(4), 737-740.

Scheme 26 (1 Reaction) Steps: **1** Yield: **90%**



Relative stereochemistry shown

Supplier (1)

31-535-CAS-7192040 Steps: **1** Yield: **90%**

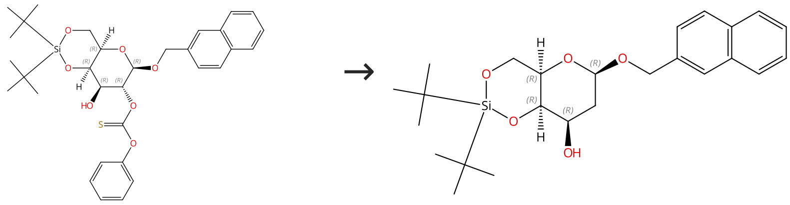
1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 4 h, reflux

Radical induced fragmentation in a benzoxocane ring system: application to the synthesis of elvirol and a formal synthesis of 7,8-dihydroxycalamenene

By: Roy, Amalesh; et al

Tetrahedron (2012), 68(32), 6575-6580.

Scheme 27 (1 Reaction) Steps: **1** Yield: **90%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown

31-614-CAS-32750668 Steps: **1** Yield: **90%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Benzene; 2 h, 80 °C

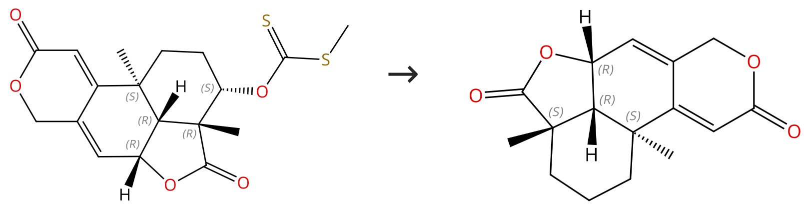
Experimental Protocols

Synthetic Modifications of the Linkage Region of Proteog lycans and Impact on CSGalNAcT-1

By: Ayela, Benjamin; et al

European Journal of Organic Chemistry (2022), 2022(2), e202101422.

Scheme 28 (1 Reaction) Steps: **1** Yield: **90%**



Absolute stereochemistry shown

Absolute stereochemistry shown, Rotation (-)

Supplier (1)

31-614-CAS-35889785 Steps: **1** Yield: **90%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 1.5 h, rt; rt → 150 °C; 12 h, 150 °C

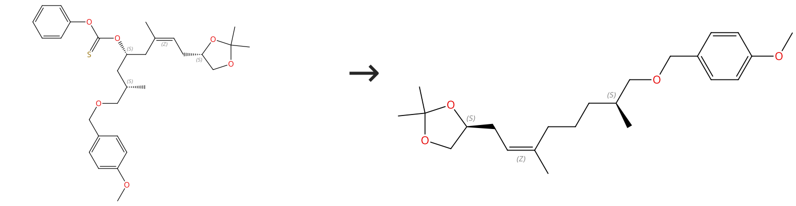
Experimental Protocols

Total syntheses of wentilactones A and B, and related nordit erpene dilactones

By: Hou, Wu; et al

Chemical Communications (Cambridge, United Kingdom) (2022), 58(89), 12487-12490.

Scheme 29 (1 Reaction) Steps: **1** Yield: **90%**



Absolute stereochemistry shown
Double bond geometry shown

Absolute stereochemistry shown,
Rotation (+)
Double bond geometry shown

31-535-CAS-5928091 Steps: **1** Yield: **90%**

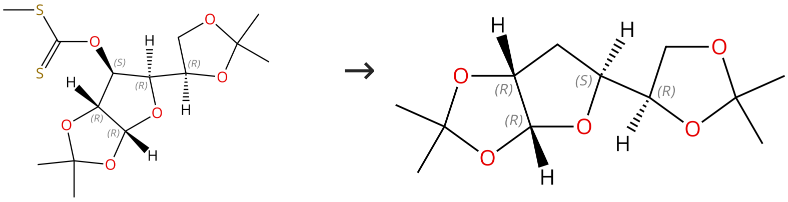
1.1 **Reagents:** Azobisisobutyronitrile, Tributylstannane

Total syntheses of epothilones B and D: applications of allylstannanes in organic synthesis

By: Martin, Nathaniel; et al

Tetrahedron Letters (2001), 42(47), 8373-8377.

Scheme 30 (1 Reaction) Steps: **1** Yield: **90%**



Absolute stereochemistry shown,
Rotation (-)

Suppliers (30)

Absolute stereochemistry shown,
Rotation (-)

Suppliers (20)

31-535-CAS-12674528 Steps: **1** Yield: **90%**

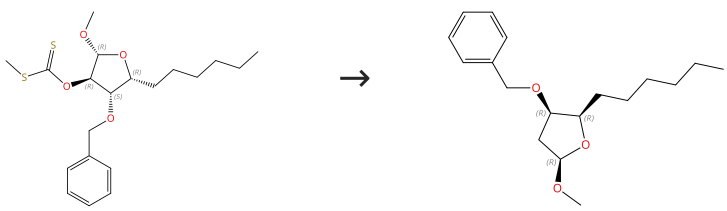
1.1 **Reagents:** Triphenylsilane
Catalysts: Di-*tert*-butyl hyponitrite
Solvents: 1,4-Dioxane

Formation of silanethiols by reaction of silanes with carbonyl sulfide: implications for radical-chain reduction of thiocarbonyl compounds by silanes

By: Cai, Y.; et al

Tetrahedron Letters (2001), 42(4), 763-766.

Scheme 31 (1 Reaction) Steps: **1** Yield: **89%**



Absolute stereochemistry shown,
Rotation (-)

Absolute stereochemistry shown,
Rotation (-)

31-614-CAS-34076190 Steps: **1** Yield: **89%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 2 h, 110 °C

Experimental Protocols

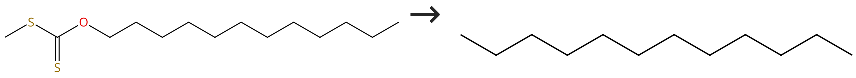
Study of the β-oxygen effect in the Barton-McCombie reaction for the total synthesis of (4R,5R)-4-hydroxy-γ-decalactone (Japanese orange fly lactone): a carbohydrate based approach

By: Desireddi, Janardana Reddi; et al

RSC Advances (2022), 12(39), 25520-25527.

Scheme 32 (1 Reaction)

Steps: **1** Yield: **89%**



Suppliers (3)

Suppliers (111)

31-533-CAS-165739

Steps: **1** Yield: **89%**

1.1 **Reagents:** Azobisisobutyronitrile, Tris(trimethylsilyl)silane
Solvents: Toluene; 5 min, 130 °C

Experimental Protocols

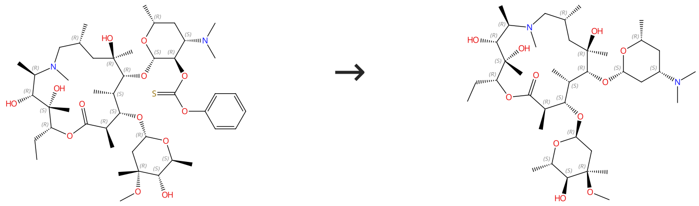
Safe, facile radical-based reduction and hydrosilylation reactions in a microreactor using tris(trimethylsilyl)silane

By: Odedra, Arjan; et al

Chemical Communications (Cambridge, United Kingdom) (2008), (26), 3025-3027.

Scheme 33 (1 Reaction)

Steps: **1** Yield: **89%**



Absolute stereochemistry shown

Absolute stereochemistry shown

31-614-CAS-31480341

Steps: **1** Yield: **89%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 3 h, rt → 90 °C

Experimental Protocols

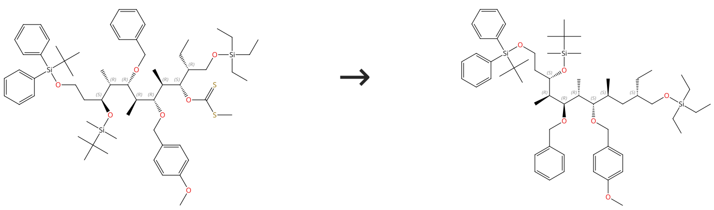
Unprecedented Epimerization of an Azithromycin Analogue: Synthesis, Structure and Biological Activity of 2'-Dehydroxy-5''-Epi-Azithromycin

By: Kragol, Goran; et al

Molecules (2022), 27(3), 1034.

Scheme 34 (1 Reaction)

Steps: **1** Yield: **87%**



Absolute stereochemistry shown, Rotation (-)

Absolute stereochemistry shown, Rotation (-)

31-535-CAS-5295815

Steps: **1** Yield: **87%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 8 h, rt → reflux

Experimental Protocols

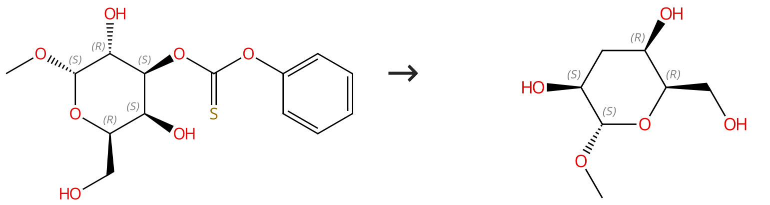
Stereoselective synthesis of the C31-C41 spirolactam fragment of Sanglifehrin A

By: Yadav, J. S.; et al

European Journal of Organic Chemistry (2013), 2013(14), 2849-2858.

Scheme 35 (1 Reaction)

Steps: **1** Yield: **87%**



Absolute stereochemistry shown

Supplier (1)

Absolute stereochemistry shown, Rotation (-)

Supplier (1)

31-535-CAS-12452649

Steps: **1** Yield: **87%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 4 h, reflux

Experimental Protocols

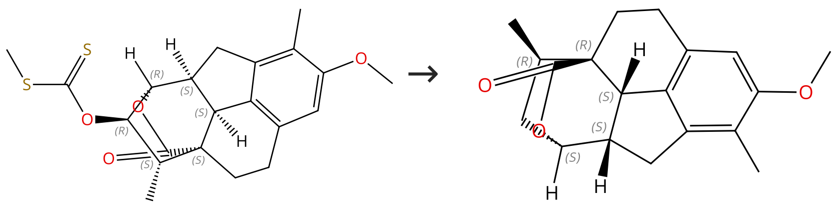
Organotin-Catalyzed Highly Regioselective Thiocarbonylation of Non-Protected Carbohydrates and Synthesis of Deoxy Carbohydrates in a Minimum Number of Steps

By: Muramatsu, Wataru; et al

Chemistry - A European Journal (2012), 18(16), 4850-4853, S4850/1-S4850/63.

Scheme 36 (1 Reaction)

Steps: **1** Yield: **86%**



Absolute stereochemistry shown, Rotation (-)

Absolute stereochemistry shown

31-535-CAS-23370985

Steps: **1** Yield: **86%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 1.5 h, 95 °C

Experimental Protocols

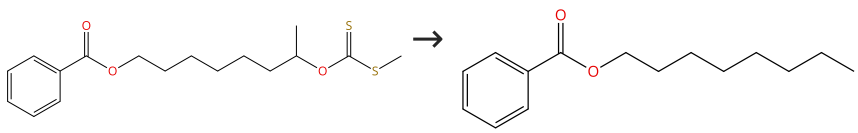
Asymmetric total synthesis of cephanolide B

By: Zhang, Hongyuan; et al

Organic Chemistry Frontiers (2021), 8(3), 555-559.

Scheme 37 (1 Reaction)

Steps: **1** Yield: **86%**



Suppliers (56)

31-535-CAS-22852302

Steps: **1** Yield: **86%**

1.1 **Reagents:** Tris(trimethylsilyl)silane
Catalysts: Chlorophyll a
Solvents: Dimethyl sulfoxide; 5 h, 50 - 55 °C

Experimental Protocols

Red-Light-Mediated Barton-McCombie Reaction

By: Ogura, Akihiro; et al

Bulletin of the Chemical Society of Japan (2020), 93(7), 936-941.

Scheme 38 (1 Reaction)

Steps: **1** Yield: **85%**

Absolute stereochemistry shown → Absolute stereochemistry shown

31-535-CAS-4588285

Steps: **1** Yield: **85%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene

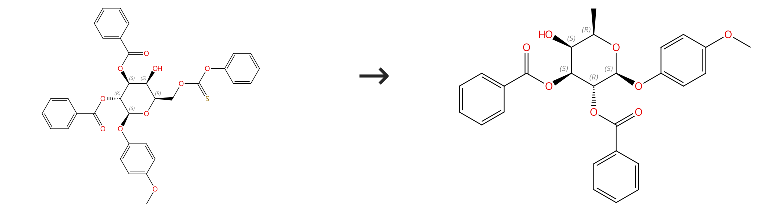
A facile synthesis of 7,10-dideoxytaxol and 7-epi-10-deoxytaxol

By: Chen, Shu Hui; et al

Tetrahedron Letters (1993), 34(43), 6845-8.

Scheme 39 (1 Reaction)

Steps: **1** Yield: **85%**



Absolute stereochemistry shown, Rotation (+) → Absolute stereochemistry shown, Rotation (+)

31-533-CAS-22488201

Steps: **1** Yield: **85%**

1.1 **Reagents:** Tris(trimethylsilyl)silane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 0 °C; 1 h, 80 °C

Experimental Protocols

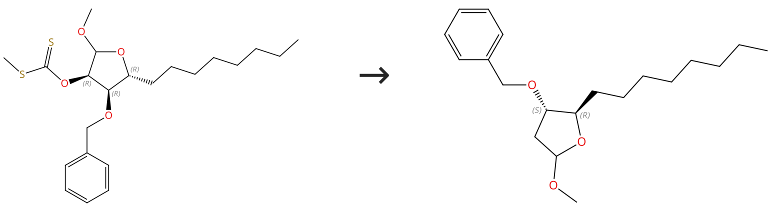
Indirect synthetic route to α-L-fucosides via highly stereose lective construction of α-L-galactosides followed by C6-deoxygenation

By: Tomida, Hirotaka; et al

Organic & Biomolecular Chemistry (2020), 18(26), 5017-5033.

Scheme 40 (1 Reaction)

Steps: **1** Yield: **85%**



Absolute stereochemistry shown → Absolute stereochemistry shown

31-614-CAS-33719120

Steps: **1** Yield: **85%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 110 °C; 1 h, 110 °C

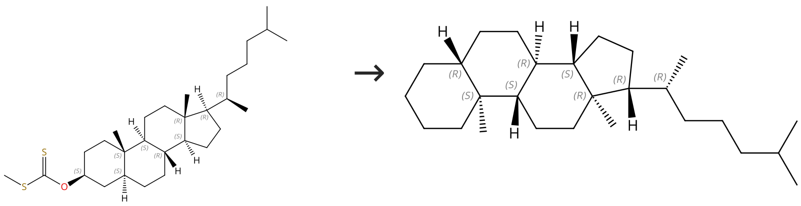
Experimental Protocols

Stereoselective Total Synthesis of 4-(S)-Hydroxy-5-(R)-octyldihydrofuran-2-one: Harvestmen Lactone

By: Desireddi, Janardana Reddi; et al

ChemistrySelect (2022), 7(27), e202201317.

Scheme 41 (1 Reaction) — Steps: **1** Yield: **85%**



Absolute stereochemistry shown — Suppliers (2)

Absolute stereochemistry shown — Suppliers (58)

31-535-CAS-10834919 — Steps: **1** Yield: **85%**

1.1 **Reagents:** Azobisisobutyronitrile, Tris(trimethylsilyl)silane
Solvents: Toluene; 5 min, 130 °C

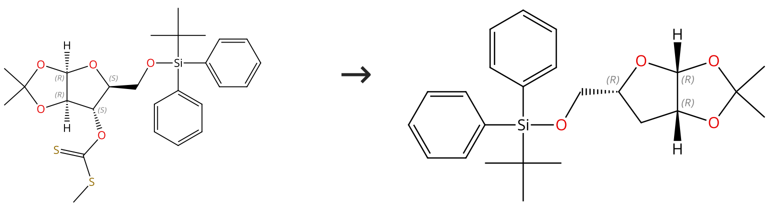
Experimental Protocols

Safe, facile radical-based reduction and hydrosilylation reactions in a microreactor using tris(trimethylsilyl)silane

By: Odedra, Arjan; et al

Chemical Communications (Cambridge, United Kingdom) (2008), (26), 3025-3027.

Scheme 42 (1 Reaction) — Steps: **1** Yield: **84%**



Absolute stereochemistry shown

Absolute stereochemistry shown, Rotation (-)

31-535-CAS-13293519 — Steps: **1** Yield: **84%**

1.1 **Reagents:** Azobisisobutyronitrile, Tributylstannane
Solvents: Toluene

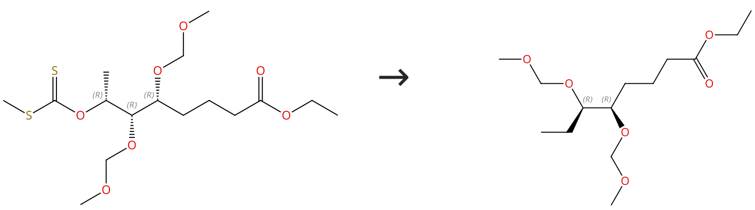
Experimental Protocols

Carbohydrate-Based Synthesis of Naturally Occurring Marine Metabolites Slagenins B and C

By: Gurjar, Mukund K.; et al

Organic Letters (2002), 4(21), 3569-3570.

Scheme 43 (1 Reaction) — Steps: **1** Yield: **84%**



Absolute stereochemistry shown, Rotation (+)

Absolute stereochemistry shown, Rotation (+)

31-535-CAS-9970826 — Steps: **1** Yield: **84%**

1.1 **Reagents:** Tributylstannane
Catalysts: Azobisisobutyronitrile
Solvents: Toluene; 30 min, reflux

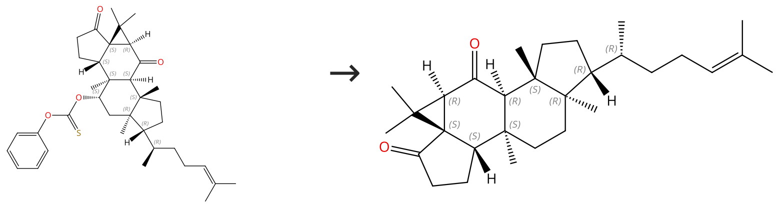
Experimental Protocols

Stereoselective synthesis of (-)-(1R,1'R,5'R,7'R)-1-hydroxy-exo-brevicomin and (+)-exo-brevicomin from 3,4,6-tri-O-acetyl-D-glucal

By: Sabitha, Gowravaram; et al

Helvetica Chimica Acta (2013), 96(8), 1610-1615.

Scheme 44 (1 Reaction) Steps: **1** Yield: **83%**



Absolute stereochemistry shown, Rotation (-) → Absolute stereochemistry shown, Rotation (-)

31-614-CAS-44831065 Steps: **1** Yield: **83%**

1.1 **Reagents:** Tributylstannane, Red 21
Solvents: Acetonitrile; 12 h, rt

1.2 **Reagents:** Water; rt

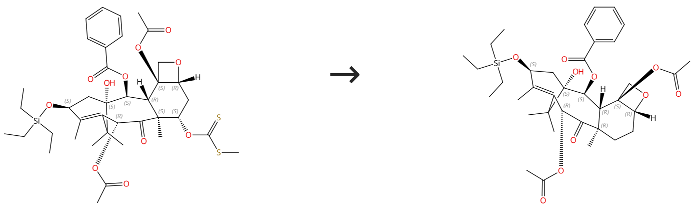
Experimental Protocols

Bioinspired Synthesis of Cucurbalsaminones B and C

By: Zhang, Mengqing; et al

Angewandte Chemie, International Edition (2025), 64(4), e202417318.

Scheme 45 (1 Reaction) Steps: **1** Yield: **83%**



Absolute stereochemistry shown → Absolute stereochemistry shown

31-535-CAS-14140643 Steps: **1** Yield: **83%**

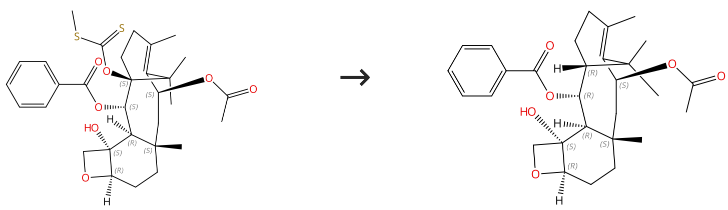
1.1 **Reagents:** Tributylstannane
Solvents: Toluene

Synthesis of 7-deoxy- and 7,10-dideoxytaxol via radical intermediates

By: Chen, Shu Hui; et al

Journal of Organic Chemistry (1993), 58(19), 5028-9.

Scheme 46 (1 Reaction) Steps: **1** Yield: **82%**



Absolute stereochemistry shown → Absolute stereochemistry shown, Rotation (+)

31-533-CAS-9899671 Steps: **1** Yield: **82%**

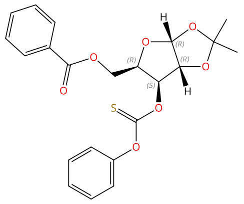
1.1 **Reagents:** Triethylamine, Hypophosphorous acid
Catalysts: Azobisisobutyronitrile
Solvents: 1,4-Dioxane; reflux

A new deoxygenation method for taxanes using hypophos phorous acid

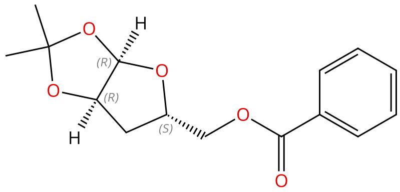
By: Gu, Jun; et al

Chinese Chemical Letters (2004), 15(6), 649-651.

Scheme 47 (1 Reaction) — Steps: **1** Yield: **82%**



Absolute stereochemistry shown

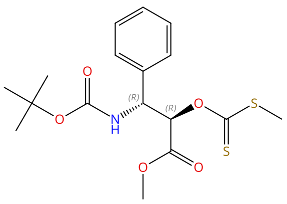


Absolute stereochemistry shown

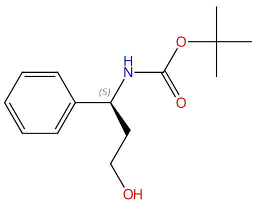
Suppliers (4)

31-535-CAS-4480328 — Steps: **1** Yield: **82%**	Total synthesis of cordycepin
1.1 **Catalysts:** Azobisisobutyronitrile **Solvents:** Toluene; 20 min, rt 1.2 **Reagents:** Tributylstannane; 2 h, reflux	By: Li, Qihuan; et al Youji Huaxue (2013), 33(6), 1340-1344.

Scheme 48 (1 Reaction) — Steps: **1** Yield: **81%**



Absolute stereochemistry shown, Rotation (-)

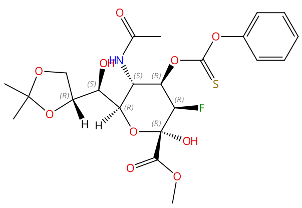


Absolute stereochemistry shown, Rotation (-)

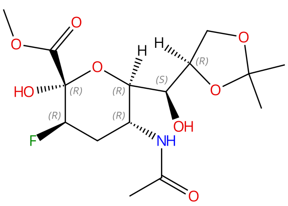
Suppliers (58)

31-515-CAS-13637392 — Steps: **1** Yield: **81%**	Enantioselective synthesis of (S)-dapoxetine
1.1 **Solvents:** Tetrahydrofuran; rt → 0 °C 1.2 **Reagents:** Lithium aluminum hydride; 2 h, reflux; reflux → 0 °C 1.3 **Solvents:** Water; cooled Experimental Protocols	By: Siddiqui, Shafi A.; et al Tetrahedron: Asymmetry (2007), 18(17), 2099-2103.

Scheme 49 (1 Reaction) — Steps: **1** Yield: **81%**



Absolute stereochemistry shown, Rotation (-)



Absolute stereochemistry shown, Rotation (-)

31-535-CAS-14335666 — Steps: **1** Yield: **81%**	The synthesis of a series of deoxygenated 2,3-difluoro-N-acteylneuraminic acid derivatives as potential sialidase inhibitors
1.1 **Reagents:** Tributylstannane **Catalysts:** 2,2-Bis(*tert*-butylperoxy)butane **Solvents:** 1,4-Dioxane; 2 h, 100 °C; 2 h, 100 °C Experimental Protocols	By: Hader, Stefan; et al Carbohydrate Research (2013), 374, 23-28.